Challenge: Payload Transport

Points: 250 • Estimated Time: 90 Minutes

Challenge Scope

The robot must carry a 500-gram weight across a fragile cardboard bridge without causing structural collapse.

Instructions & Engineering Hints

• **Objective:** Design and program your robot to securely transport a 500-gram payload across a delicate cardboard bridge without causing a structural collapse.

• **Step 1 - Payload Integration:** Securely mount the 500-gram weight to your robot. Consider how the height of the center of gravity impacts the robot's stability, especially when accelerating, decelerating, or turning.

• **Step 2 - Weight Distribution:** Think about how the robot's weight is distributed across its wheels or tracks. Spreading the contact points or using continuous treads can help reduce the localized pressure exerted on the fragile bridge surface.

• **Step 3 - Navigation & Speed Control:** Program your robot to approach and traverse the bridge with precision. Sudden stops, sharp turns, or excessive speed can destabilize both the payload and the bridge structure. Implement smooth acceleration and deceleration profiles in your code.

• **Step 4 - Sensor Strategies:** Consider utilizing distance or color sensors to detect the approach and exit points of the bridge, allowing the robot to autonomously trigger its 'bridge-crossing mode' safely.